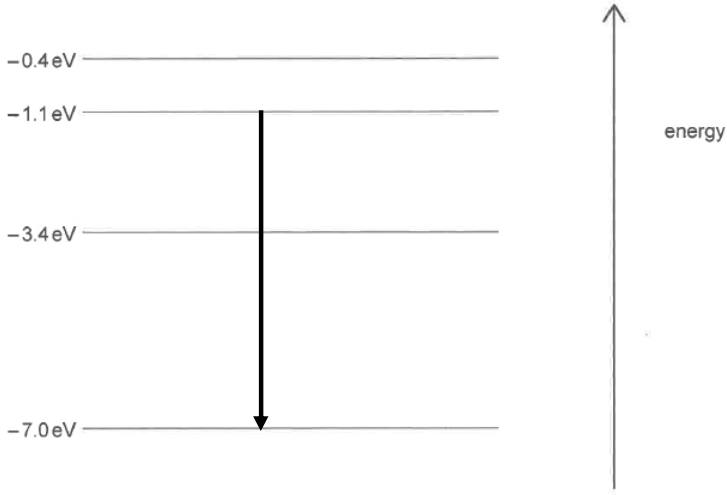


9(a)	In particulate nature, light is considered as discrete packets of energy called photon. Each photon has energy $E = hf$. Since one photon can only interact with one electron, the photons caused emission of photoelectrons whose K.E. depends on the energy provide by each photon. Since the frequency f in observation 1 is constant, the energy of each photon is constant and thus energy supplied to each photoelectron is constant. It does not depend on the intensity of the incident radiation as intensity only affect the number of photons incident per unit time on the metal and not the energy of each photons. The first observation does not support wave nature of electromagnetic radiation as wave theory states that at higher intensity, the higher power per unit area should cause photoelectrons of higher K.E. to be emitted, which was not observed in experiments. For observation 2, if frequency of the photon is below a threshold frequency, energy of each photon is not enough to remove a electron from the metal surface as its energy is lower than the minimum energy needed. This again does not support wave nature of electromagnetic radiation as with higher intensity, you expects the electrons in the metal to sooner or later accumulate sufficient energy over time to get the minimum energy needed to be emitted. Again, this was not observed in experiments.
9(b)	Threshold frequency.

No.	Solution
9(c)	$KE_{max} = hf - W_{min}$ $= \frac{hc}{\lambda} - \phi$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{210 \times 10^{-9}} - (1.60 \times 10^{-19})(4.33)$ $= 2.54 \times 10^{-19} \text{ J}$
9(d)	$KE_{max} = eV_s$ $V_s = \frac{KE_{max}}{e}$ $= \frac{2.54 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 1.59 \text{ V}$
9(e)(i), (ii), (iii)	(e)(ii) shorter wavelength means photons have higher frequency. Thus the incoming photons have higher energy resulting in emitted electrons having higher K.E. and thus higher stopping potential. Since intensity is the same, number of photons emitted per unit time by the source is lower resulting in lower number of photoelectrons emitted and thus a lower current. (e)(iii) with greater work function energy, the maximum K.E. of emitted electrons is lower and thus, the stopping potential is lower.
9(e)(iv)	The photoelectrons are emitted with varying K.E. At V values equal to or greater than the stopping potential, the electrode is negative with respect to the zinc plate and at these potentials, photoelectrons emitted with the maximum K.E. possible will have insufficient K.E. to reach the electrode as all the K.E. is already converted to electric potential energies before reaching the electrode. Since no electrons reaches the electrode and current is the rate of charge flow, the current is zero.

No.	Solution
9(f)(i)	 Energy of the 210 nm electromagnetic radiation from (c) is $E = \frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{210 \times 10^{-9}}$ $= 9.47 \times 10^{-19} \text{ J}$ $= 5.92 \text{ eV}$ Thus, this will be due to emission between energy level -1.1 eV to -7.0 eV.
9(f)(ii)	Longest wavelength emitted through transition with the smallest energy difference. $(-0.4 - (-1.1))(1.6 \times 10^{-19}) = \frac{hc}{\lambda}$ $\lambda = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(-0.4 - (-1.1))(1.6 \times 10^{-19})}$ $= 1.78 \times 10^{-6} \text{ m}$